

Software Project Management (SPM) (CSC415)

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Chapter 5: Resource Allocation

Resource

- In project management, a resource is defined as anything needed to complete a project successfully.
- This includes raw materials, operational supplies, funding, personnel, technology, and any other items or services required along the way.
- Resources can be project specific, or they can be general and used across multiple projects.
- It's important to accurately assess all of the resources necessary for the successful completion of a project plan and deliverables before work begins.

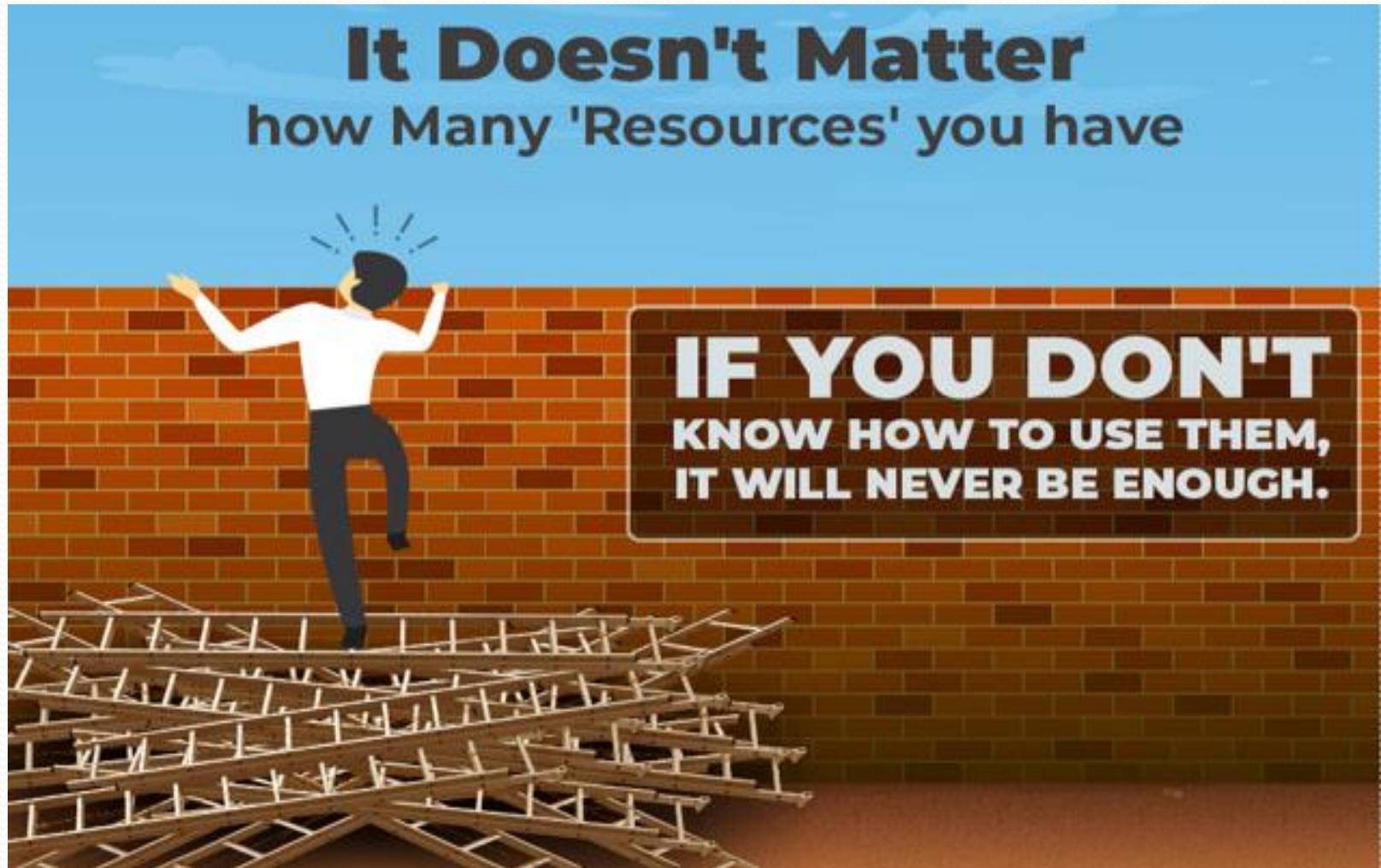
Resource

- Without sufficient resources allocated across each stage of the development process, a project will ultimately fail.
- So appropriate budgeting and allocations are made for each aspect of the activities associated with the project.

Types of Resource

- All organizations have at least four types of resources (or assets) that can be used to achieve desired objectives:
 - *Financial Resources,*
 - *Physical Resources,*
 - *Human Resources, and*
 - *Technological Resources.*
- ***Intangible Resources:*** Goodwill, Brand Equity, Intellectual Property, Licensing and Rights, Customer Lists, Research and Developments etc.

Resource Allocation



Resource Allocation

- After the activities have been identified using various techniques and tabulated into a Work-Break-Down the resources need to be allocated to complete the identified tasks.
- Resource Allocation is the process of assigning and scheduling available resources in the most effective and economical way possible.
- Projects always need resources but they can often be scarce.

Resource Allocation

- The task, therefore, lies with the project manager to determine the proper timing and allocation of those resources within the project schedule.
- It is the management and delegation of resources throughout a project to ensure that it runs as smoothly and successfully as possible.
- Goal of resource allocation is to optimize use of limited supply that requires making trade-offs among:
 - *Time Constrain*
 - *Resource Constrain*

Resource Allocation

- Resource allocation can be used as a control mechanism of the project as well.
- Allocating more, less, or different physical resources can affect the progress of activities, and can affect their cost and/or the quality of outputs/deliverables as well.
- The final result of resource allocation will normally be a number of schedules including:
 - **Activity Schedule:** *indicating start and completion dates for each activity*
 - **Resource Schedule:** *indicating dates when resources are needed and level of resources*
 - **Cost Schedule:** *shows accumulative expenditures.*

Issues in Resource Allocation

- **Availability:**
 - We need to know of a particular individual available when required.
- **Criticality:**
 - Allocate the more experienced person to the critical activity, which may be tough task to do.
- **Risk:**
 - Allocate the highly experienced personnel to highest risk activity to reduce overall project uncertainty.
- **Training:**
 - Proper training need to be provided to every person about the task they are given to.
- **Team Building:**
 - Proper selection of members for particular team to do particular task.

Challenges of resource allocation:

- Changes in project
- A lack of qualified resources
- Poor resource capacity measuring
- Changes in resource availability
- Not assessed risks
- Task delays
- Working with international teams



Resource Organization

- A program organization chart is essential to allocate staff effectively,
 - Develop the hierarchical program organization.
 - Identify Roles and Responsibilities.
 - Plan for number of staff in each role (at a high level).
 - Establish Teams.

Identifying Resource Requirement

- Resource requirements in project management allow project managers to know what they'll need to execute their tasks and deliver the project on time and within their budget.
- Knowing one's resource requirements will also help project managers to get the most out of their resources.
- Resource requirements differ from project to project, of course, but also on how productive your team can be and even project switching, which is as team members pass tasks to one another.

Identifying Resource Requirement

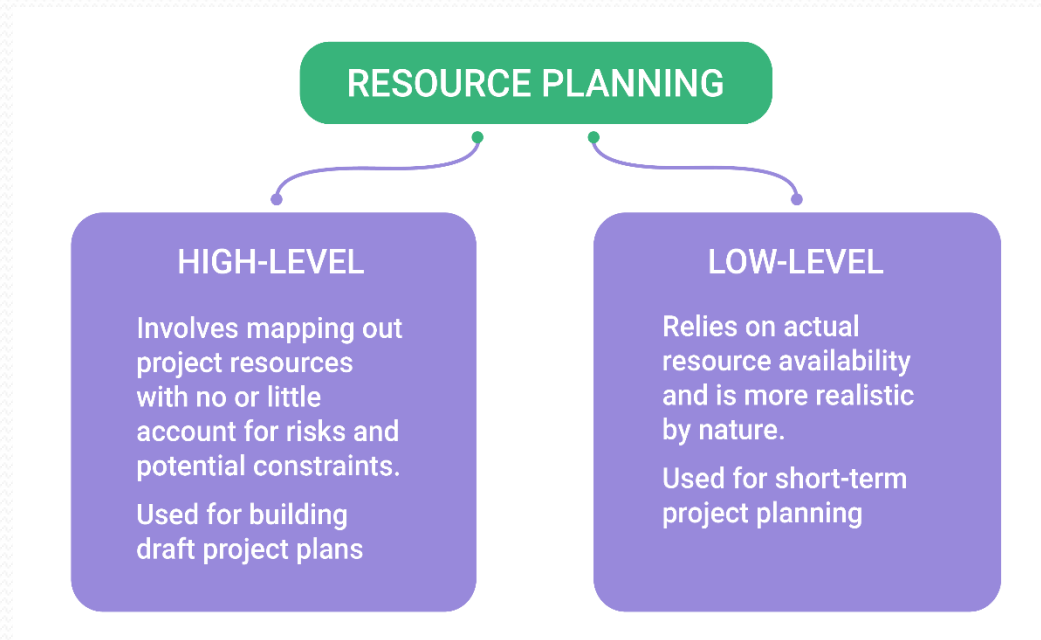
- There are some steps that can help to narrow down the resource requirement and to forecast as accurate as possible:
 - *Establish Project Scope*
 - *Make Activity Break Down Structure*
 - *Estimate Task Duration*
 - *Identify Requirements of each Task*
 - *Create Resource Schedule*
 - *Assign Resources to Tasks*
 - *Use Resource Tracking Tools*

Identifying Resource Requirement

- *Example for Activity :- Install Network Hardware for 20 computers.*
 - Work units - 20.
 - Basic Skill – Bachelors Degree in related field.
 - Task Complexity: 5.
 - Task Category: Skilled (other categories may be Management, Leadership, Expert)

Resource Planning

- Lack of project resources can become a constraint on the completion of project activities, therefore proper resource planning and scheduling are key to successful project management.



Resource Planning

- Benefits of proper resource planning include:
 - *Proper skills and talent utilization ensuring better productivity and job satisfaction*
 - *Easier and more accurate capacity tracking*
 - *Simple project progress tracking*
 - *Better team collaboration and productivity*
 - *Improved job satisfaction and retention in your team*
 - *Stop team burnout*

Resource Scheduling

- After all the required resources have been identified, they need to be scheduled effectively.
- The earliest start dates, last start dates will need to be taken into account to schedule resources efficiently.
- Resources should be balanced throughout the project.
- In contrast, strong resource scheduling brings the following benefits:
 - *Easier project progress monitoring*
 - *Optimize resource schedules anytime*

Resource Scheduling

- It should always be in the project planner's mind, right from the start of the project.
- During the resource scheduling and allocation phase of the planning activity a lot of the plan will change.
- Most of the issues with respect to resource allocation and scheduling arise after the project starts (normally after about 30% of the activities are complete).

Resource Scheduling

- ***Tools for Project Scheduling:***
 - **PERT Chart:**
 - A graphical representation of a project's schedule
 - Shows the sequence of tasks, which tasks can be performed simultaneously
 - **Gantt Chart:**
 - Illustrates the start and finish dates of the terminal elements and summary elements of a project.
- **Work Breakdown Structure:**
 - What are tasks to be done to complete the project

Resource Balancing

- Resource balancing is a critical aspect of project management that involves effectively allocating and managing project resources to achieve project goals and objectives.
- It ensures that the right resources are available at the right time, in the right quantities, and at the cost to support project activities and deliverables.
- Resource balancing helps project managers optimize resource utilization, minimize waste, and maintain project quality and schedule.

Resource Balancing

- For equally allocating the task and resources among the team members:
 - First we make sure, that the Work Breakdown Structure (WBS) is through and accurate and list all resources and activities.
 - Now, we identify most critical tasks and allocate best resources for that work.

Resource Balancing

- Then, we begin resource leveling as:
 - Check to see how many hours are available per team member for work then determine how many hours have been allocated for each person.
 - If there is any uncertainty in allocation of time/person then the tasks should be adjusted so that the no. of required work hours are equivalent to the no. of hours available.

Resource Leveling

- In many cases projects don't always go as planned.
- You may encounter resource shortages and have to find ways to adapt to ensure project deliverables aren't affected.
- That's where resource leveling comes in.
- It allows you to re-adjust timelines so that a project can be finished with the available resources.
- With this technique, you can allocate resources accordingly to ensure goals and objectives are met.

Resource Smoothing

- Resource smoothing is defined as a technique that adjusts the activity of a schedule model so that all the requirements for the resource do not go beyond limits than that of pre-defined planning.
- It is one of the project management tools used in the resource optimization techniques.
- There are only few reusable resources that are limitless, thus the time schedules have to be imposed and adjusted to manage limited availability of resources in given time.

Resource Smoothing

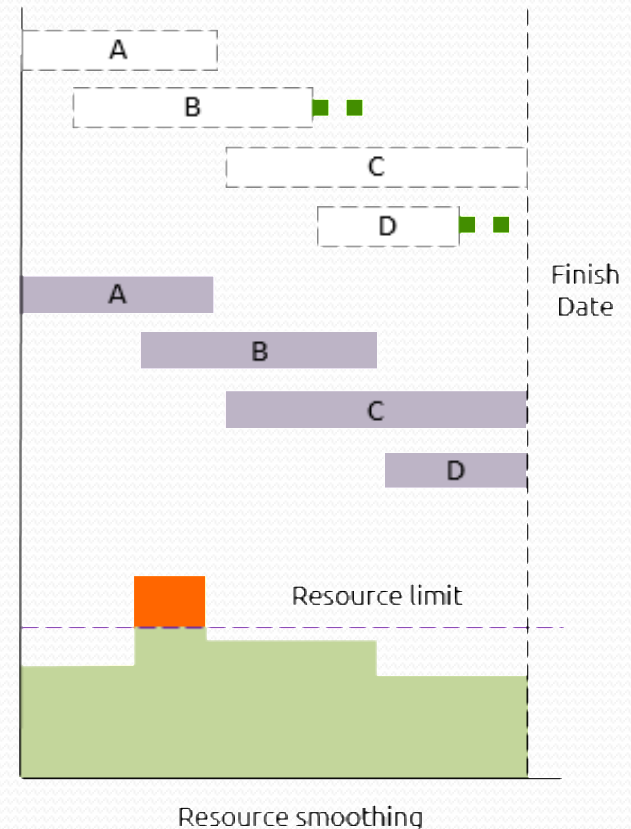
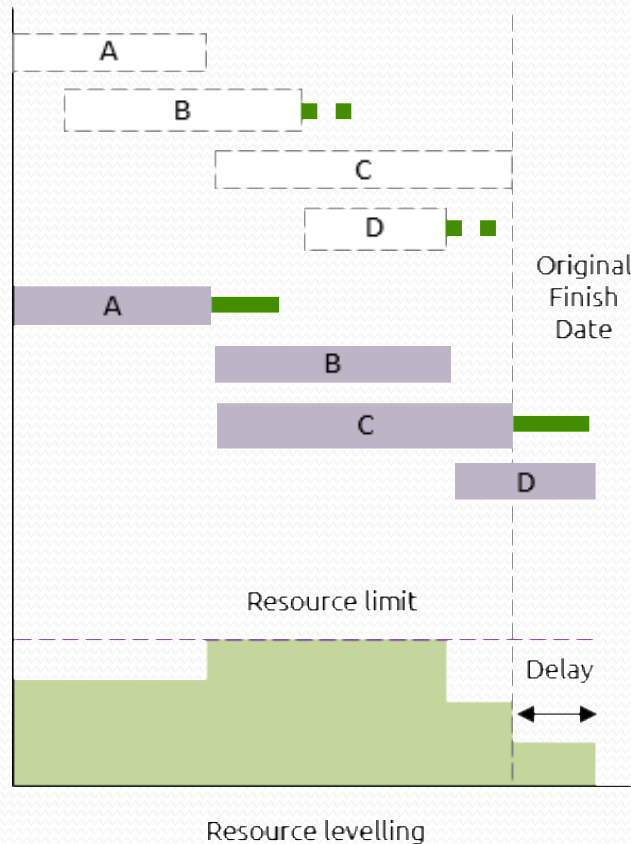
- The main objective of this tool is to complete work/activity within required date and at the same time avoiding peaks and troughs or the resource demand.

Smoothing vs Leveling

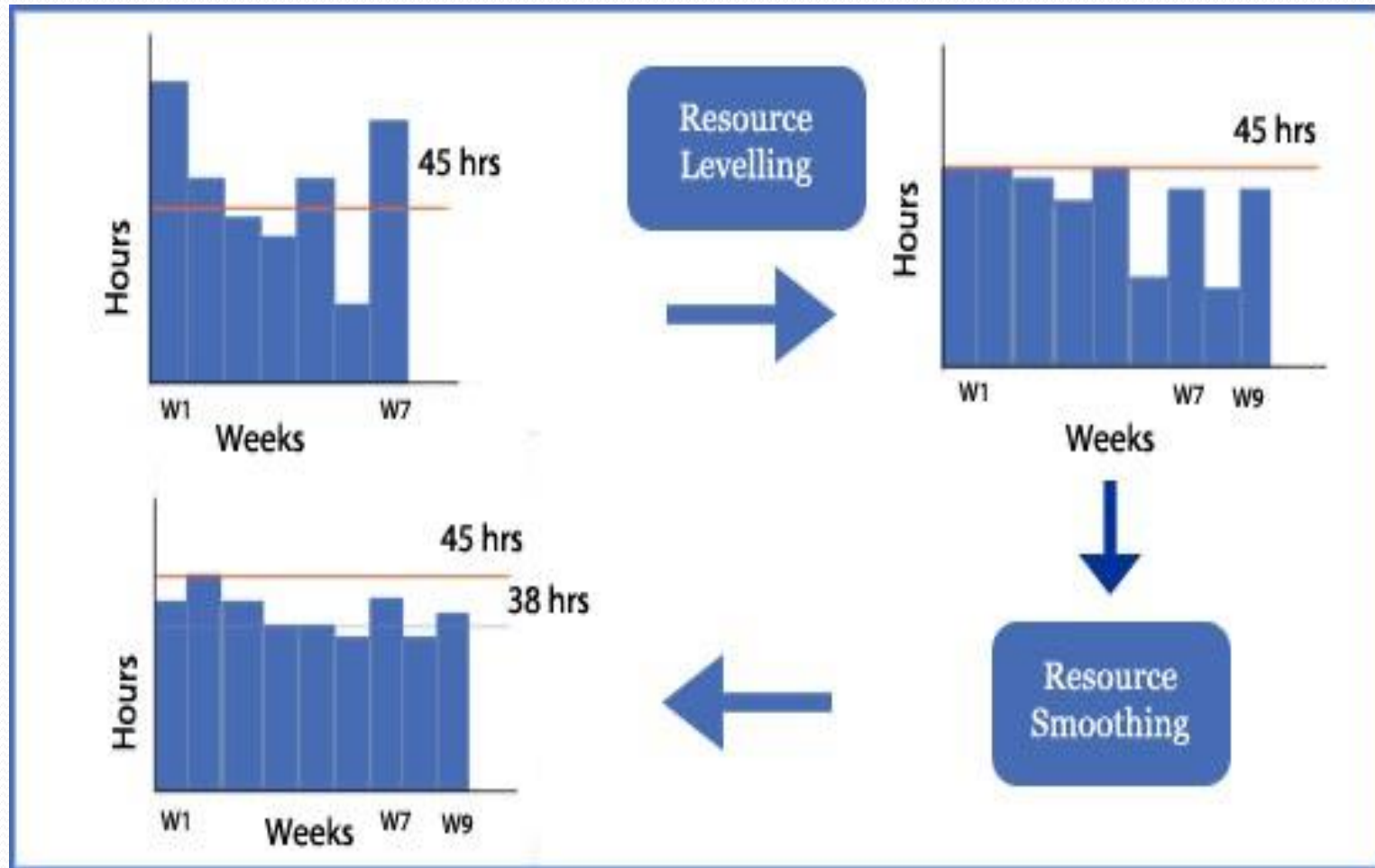
- Resource Levelling may impact your critical path i.e. Project could delay whereas Resource Smoothing doesn't.
- Resource Smoothing is performed after resource levelling to eliminate the peaks and troughs from the schedule using their total and free floats.
- Resource levelling balance the resource demands by levelling the under and over allocated critical resources.
- Resource smoothing makes sure that resources are evenly allocated

Smoothing vs Leveling

- Resource levelling answers the question of when you will be able to complete the project with the given resources.



Smoothing vs Leveling



Resource Conflict

- Sometimes a limited resource is needed (at the same time) by several activities in one project or by different activities in multiple projects (so some of the activities must wait).
- Setting priority rules for which activity get the constrained resource first can be a means of project control.
- In the case of multiple projects, the goals and importance of all the projects should be considered.



End of Chapter 5

Thank You !!!!